

PPP3: Working with Databases

Databases can be efficiently manipulated by using Python. This article gives another way to solve a problem introduced in an earlier article by using Python with databases.

In this month's column, let's look at working with databases using Python. I have illustrated the concepts using the problem of 'Project Effort summaries' that I had introduced in this column earlier ('Practical Python Programming: Interfacing with Excel', OSFY, March 2015). Let's begin with whirlwind tours of RDBMS and SQL. Readers already familiar with them are encouraged to directly go to the section 'Introducing the SQLite3 package'. Others should continue to get a bird's eye view of the basics of databases and SQL.

A problem of 'Project Effort summaries'

Assume that a group of employees in a particular team is working across different projects for different customers. The entity relationship diagram (ERD) given in Figure 1 gives an idea about the association between an employee, a project and a customer.

An employee could be working on one or more projects. A project could also have one or more employees. A project

is targeted to be delivered to only one customer (in our example). A customer may be involved with multiple projects.



Note: Data about each of these entities could be modelled in a relational table of the same name. We will discuss more about tables in a subsequent section.

Now, the management may be interested in answers to different questions based on the data available, like how do we find out the summary effort spent by the team on a particular customer over a period of time? Or, what is the actual effort spent on a project till date (and how does it compare with the planned effort)?

To begin with, a part of the solution will have to be about deciding where and how to store the data. Here, we are first talking about the persistence layer for the information collected. While Python could be used to develop a full-fledged solution with a layer managing the data along with access to it (which could be made independent of the format

